

■ 1.9.3 T_EX Formatting System

When you are preparing documents for publication, you may need to use a full-function formatting system, such as T_EX. You can use **TeXForm** to write out results from *Mathematica* in a form suitable for input to T_EX.

TeXForm[expr] write out *expr* in T_EX input form

Mathematica output for formatting systems.

Here is an expression, written out in standard mathematical form.

In[1] := (x + y)² / (x y)

$$\textit{Out}[1] = \frac{(x + y)^2}{x y}$$

Here is the expression in T_EX input form.

In[2] := **TeXForm**[%]

Out[2]//**TeXForm** =

$$\frac{\left(\left(\left(x + y\right)\right)^2\right)}{x y}$$

Here is a more complicated example.

This is the standard *Mathematica* form of an expression.

In[3] := **Sum**[**Sqrt**[8]/9801 (4*n*)! (1103 + 26390 *n*) /
 (n!⁴ 396^{4*n*}), {*n*, 1, Infinity}]}

$$\textit{Out}[3] = \text{Sum}\left[\frac{2 \sqrt{2} (4 n)! (1103 + 26390 n)}{9801 396^4 n!^4}, \{n, 1, \text{Infinity}\}\right]$$

Here is the expression in T_EX input form.

In[4] := **TeXForm**[%]

Out[4]//**TeXForm** =

$$\sum_{n=1}^{\infty} \frac{2 \sqrt{2} \left(4 n\right)! \left(1103 + 26390 n\right)}{9801 \left\{396\right\}^4 n!^4}$$

Here is the result of running the T_EX input through T_EX:

$$\sum_{n=1}^{\infty} \frac{\sqrt{8}}{9801} (4n)! \frac{1103+26390n}{n!^4 396^{4n}}$$

TeXForm automatically converts a number of *Mathematica* symbol names to T_EX forms.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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<i>Mathematica form</i>	<i>T_EX form</i>
alpha, beta, gamma, etc.	α, β, γ , etc.
ALPHA, BETA, GAMMA, etc.	A, B, Γ , etc.
I, E, Pi, Infinity	i, e, π, ∞

Some transformations made by TeXForm.